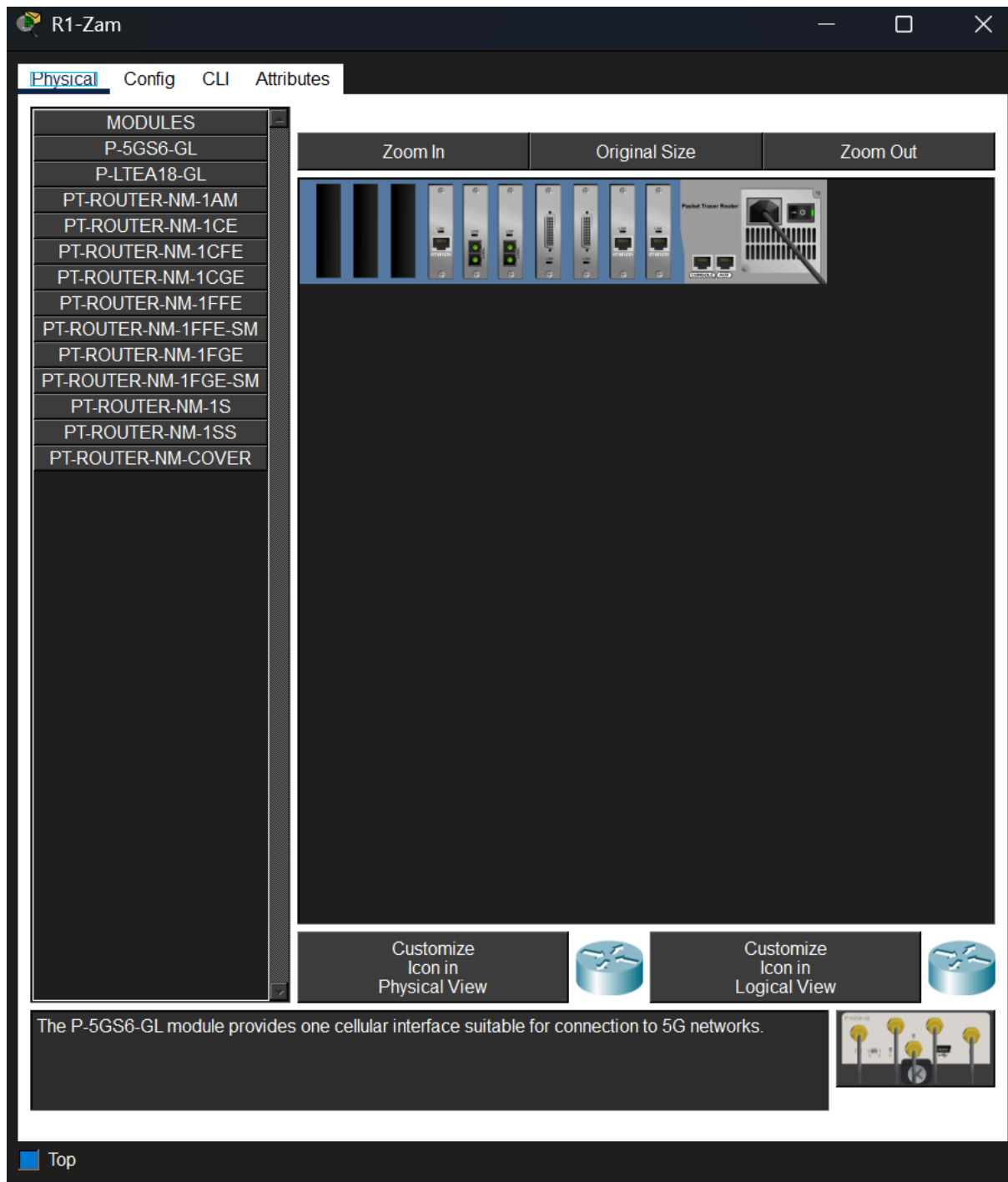


Nizamuddin Giffa Abdul Aziz Zahrani
IS ERP 2
012202400062



Description: > To accommodate multiple LAN segments within the network topology, additional physical modules were required for the routers. The **PT-ROUTER-NM-1CFE** module was manually installed into the empty slots of both the R1-Zam and R2-Zam routers. This module provides an extra Fast-Ethernet interface utilizing copper media, which is essential for connecting the routers to the department switches.

Description: > To accommodate multiple LAN segments within the network topology, additional physical modules were required for the routers. The **PT-ROUTER-NM-1CFE** module was manually installed into the empty slots of both the R1-Zam and R2-Zam routers. This module provides an extra Fast-Ethernet interface utilizing copper media, which is essential for connecting the routers to the department switches.

R2-Zam

Physical

Config

CLI

Attributes

MODULES

P-5GS6-GL

P-LTEA18-GL

PT-ROUTER-NM-1AM

PT-ROUTER-NM-1CE

PT-ROUTER-NM-1CFE

PT-ROUTER-NM-1CGE

PT-ROUTER-NM-1FFE

PT-ROUTER-NM-1FFE-SM

PT-ROUTER-NM-1FGE

PT-ROUTER-NM-1FGE-SM

PT-ROUTER-NM-1S

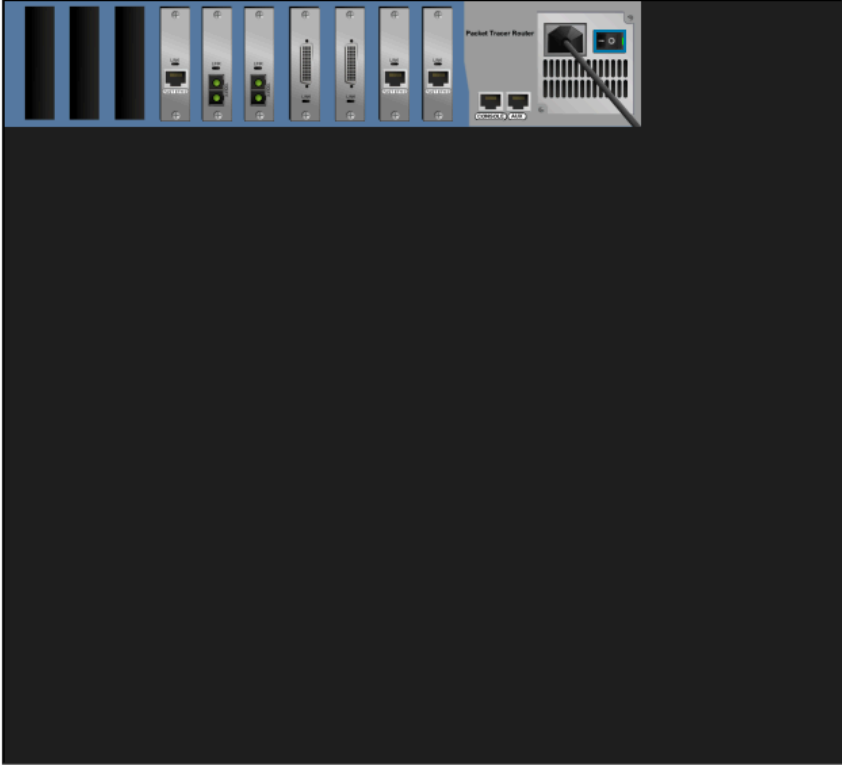
PT-ROUTER-NM-1SS

PT-ROUTER-NM-COVER

Zoom In

Original Size

Zoom Out

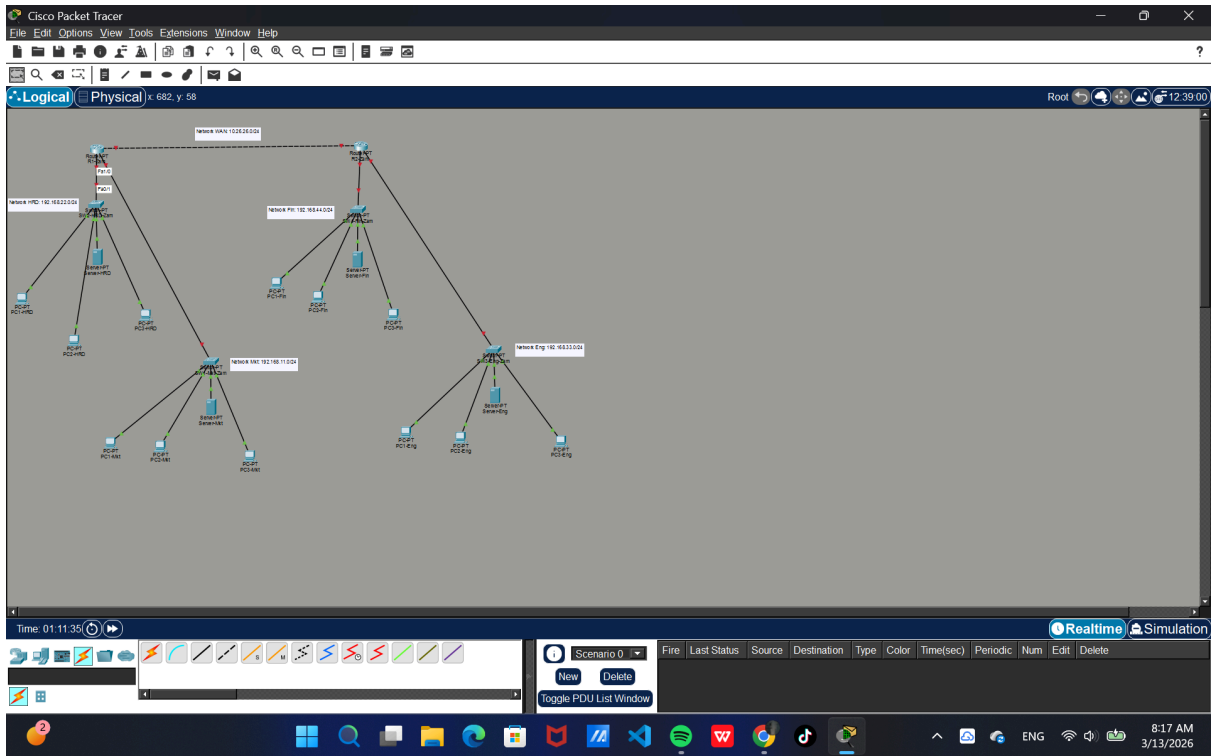
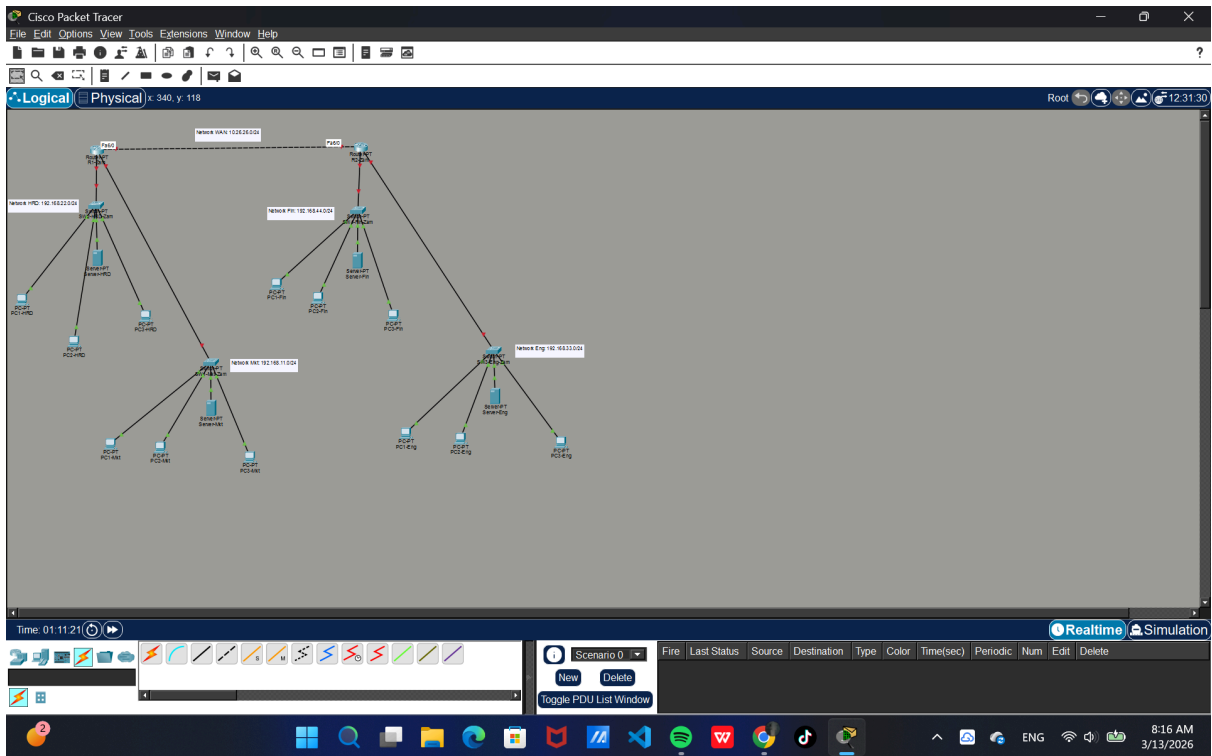


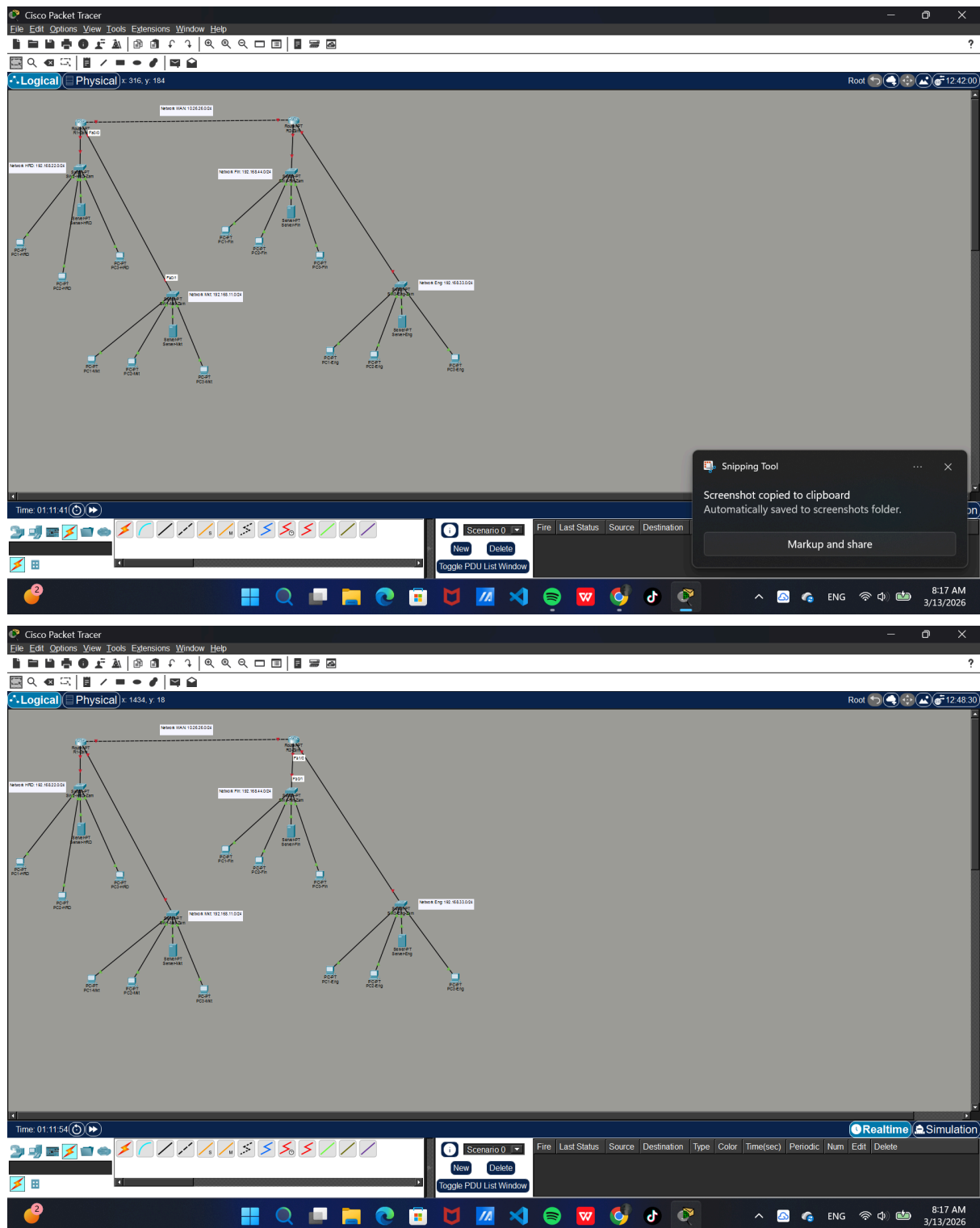
Customize Icon in Physical View

Customize Icon in Logical View

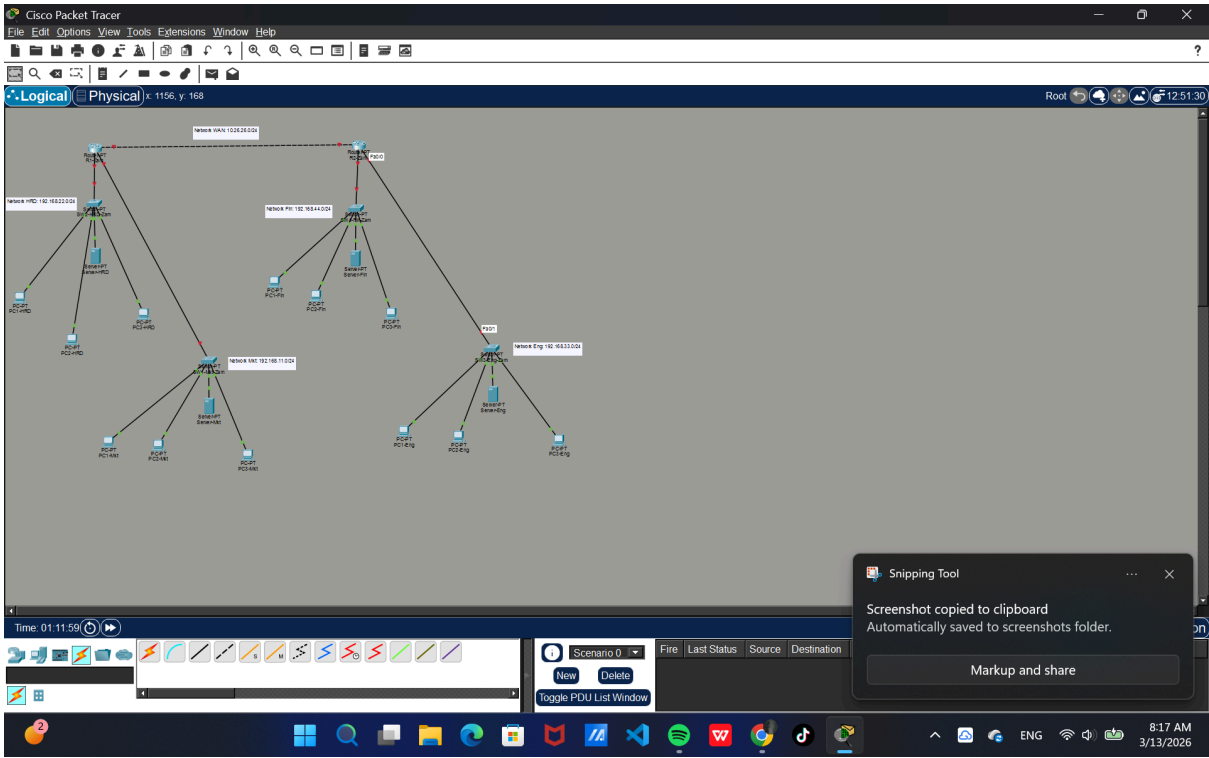
The PT-ROUTER-NM-1CFE Module provides one Fast-Ethernet interface for use with copper media. Ideal for a wide range of LAN applications, the Fast Ethernet network modules support many internetworking features and standards. Single port network modules offer autosensing 10/100BaseTX or 100BaseFX Ethernet. The TX (copper) version supports virtual LAN (VLAN) deployment.

Top





The final network topology consists of two main routers connecting four distinct departmental networks: HRD, Marketing, Finance, and Engineering. To verify the integrity of the configuration, cross-network connectivity was tested using Simple PDU (Packet Data Unit). The Realtime simulation indicates successful ICMP packet delivery between different network segments, confirming that the physical cabling and static routing protocols are functioning as intended.





The screenshot shows a terminal window titled "R1-Zam" with tabs for "Physical", "Config", "CLI", and "Attributes". The "CLI" tab is active, displaying the following text:

```
San Jose, California 95134-1700

Cisco Internetwork Operating System Software
IOS (tm) PT1000 Software (PT1000-I-M), Version 12.2(28), RELEASE SOFTWARE (fc5)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2005 by cisco Systems, Inc.
Compiled Wed 27-Apr-04 19:01 by miwang

PT 1001 (PTSC2005) processor (revision 0x200) with 60416K/5120K bytes of memory
.
Processor board ID PT0123 (0123)
PT2005 processor: part number 0, mask 01
Bridging software.
X.25 software, Version 3.0.0.
5 FastEthernet/IEEE 802.3 interface(s)
2 Low-speed serial(sync/async) network interface(s)
32K bytes of non-volatile configuration memory.
63488K bytes of ATA CompactFlash (Read/Write)

--- System Configuration Dialog ---

Would you like to enter the initial configuration dialog? [yes/no]:

Press RETURN to get started!

Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface FastEthernet0/0
Router(config-if)#ip address 10.26.26.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to
up
exit
Router(config)#
```

At the bottom right of the terminal window, there are "Copy" and "Paste" buttons. At the bottom left, there is a "Top" button.

Description: The core network routing and IP assignments were executed through the Cisco IOS Command Line Interface (CLI). IP addresses and subnet masks were assigned to the active FastEthernet interfaces and their states were changed to 'up' using the `no shutdown` command. Furthermore, DHCP Pools (such as Pool Fin and Pool ENG) were established within the router configuration to automate IP distribution. Finally, static routing was implemented to define the exact paths for data packets traveling between the R1 and R2 network gateways.

R2-Zam

PhysicalConfigCLIAttributes

```
Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to
up
exit
Router(config)#interface FastEthernet0/1
%Invalid interface type and number
Router(config)#ip address 192.168.33.1 255.255.255.0
      ^
% Invalid input detected at '^' marker.

Router(config)#no shutdown
      ^
% Invalid input detected at '^' marker.

Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface FastEthernet6/0
Router(config-if)#ip address 192.168.33.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet6/0, changed state to up
exit
Router(config)#ip dhcp pool Fin
Router(dhcp-config)#network 192.168.44.0 255.255.255.0
Router(dhcp-config)#default-router 192.168.44.1
Router(dhcp-config)#exit
Router(config)#ip dhcp ENG
      ^
% Invalid input detected at '^' marker.

Router(config)#ip dhcp pool ENG
Router(dhcp-config)#network 192.168.33.0 255.255.255.0
Router(dhcp-config)#default-router 192.168.33.1
Router(dhcp-config)#exit
Router(config)#ip route 192.168.11.0 255.255.255.0 10.26.26.1
Router(config)#ip route 192.168.22.0 255.255.255.0 10.26.26.1
Router(config)#
```

CopyPaste

Top

Server-Mkt

PhysicalConfigServicesDesktopProgrammingAttributes

IP ConfigurationX

IP Configuration

DHCP

Static

IPv4 Address169.254.108.120

Subnet Mask255.255.0.0

Default Gateway0.0.0.0

DNS Server0.0.0.0

IPv6 Configuration

Automatic

Static

IPv6 Address/

Link Local AddressFE80::260:70FF:FE51:6C78

Default Gateway

DNS Server

802.1X

Use 802.1X Security

AuthenticationMD5

Username

Password

Top

PC1-Mkt

PhysicalConfigDesktopProgrammingAttributes

IP Configuration

InterfaceFastEthernet0

IP Configuration

☒ DHCP

☒ Static

IPv4 Address

169.254.112.237

Subnet Mask

255.255.0.0

Default Gateway

0.0.0.0

DNS Server

0.0.0.0

IPv6 Configuration

☒ Automatic

☒ Static

IPv6 Address

/

Link Local Address

FE80::203:E4FF:FE36:70ED

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication

MD5

Username

Password

Top

PC2-Mkt

Physical

Config

Desktop

Programming

Attributes

IP Configuration

X

Interface

FastEthernet0

IP Configuration

DHCP

Static

IPv4 Address

169.254.230.192

Subnet Mask

255.255.0.0

Default Gateway

0.0.0.0

DNS Server

0.0.0.0

IPv6 Configuration

Automatic

Static

IPv6 Address

/

Link Local Address

FE80::201:96FF:FE79:E6C0

Default Gateway

DNS Server

802.1X

Use 802.1X Security

Authentication

MD5

Username

Password

Top

PC3-Mkt

Physical

Config

Desktop

Programming

Attributes

IP Configuration

InterfaceFastEthernet0

IP Configuration

DHCP

Static

IPv4 Address

169.254.67.117

Subnet Mask

255.255.0.0

Default Gateway

0.0.0.0

DNS Server

0.0.0.0

IPv6 Configuration

Automatic

Static

IPv6 Address

/

Link Local Address

FE80::207:ECFF:FE86:4375

Default Gateway

DNS Server

802.1X

Use 802.1X Security

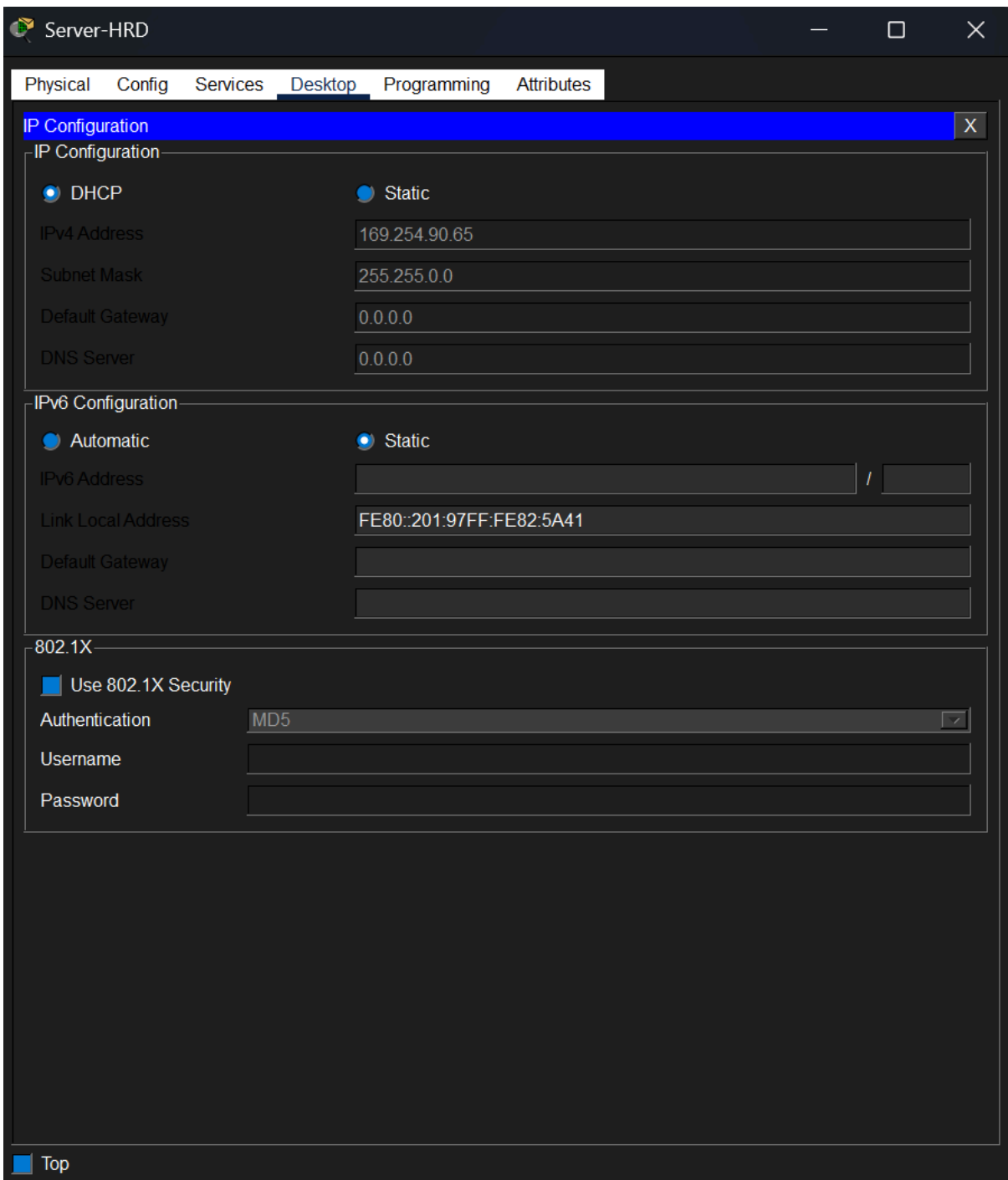
Authentication

MD5

Username

Password

Top



PC1-HRD

Physical

Config

Desktop

Programming

Attributes

IP Configuration

InterfaceFastEthernet0

IP Configuration

DHCP

Static

IPv4 Address

169.254.85.26

Subnet Mask

255.255.0.0

Default Gateway

0.0.0.0

DNS Server

0.0.0.0

IPv6 Configuration

Automatic

Static

IPv6 Address

/

Link Local Address

FE80::290:CFF:FE1C:551A

Default Gateway

DNS Server

802.1X

Use 802.1X Security

Authentication

MD5

Username

Password

Top

PC2-HRD

PhysicalConfigDesktopProgrammingAttributes

IP Configuration

X

Interface

FastEthernet0

IP Configuration

DHCP

Static

IPv4 Address

169.254.154.216

Subnet Mask

255.255.0.0

Default Gateway

0.0.0.0

DNS Server

0.0.0.0

IPv6 Configuration

Automatic

Static

IPv6 Address /

Link Local Address

FE80::20A:41FF:FEAB:9AD8

Default Gateway

DNS Server

802.1X

Use 802.1X Security

Authentication

MD5

Username

Password

Top

PC3-HRD

PhysicalConfigDesktopProgrammingAttributes

IP Configuration

InterfaceFastEthernet0

IP Configuration

☒ DHCP

☒ Static

IPv4 Address

169.254.120.145

Subnet Mask

255.255.0.0

Default Gateway

0.0.0.0

DNS Server

0.0.0.0

IPv6 Configuration

☒ Automatic

☒ Static

IPv6 Address

/

Link Local Address

FE80::202:4AFF:FEC2:7891

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication

MD5

Username

Password

Top

PC1-Eng

Physical

Config

Desktop

Programming

Attributes

IP Configuration

X

Interface

FastEthernet0

IP Configuration

DHCP

Static

IPv4 Address

192.168.44.3

Subnet Mask

255.255.255.0

Default Gateway

192.168.44.1

DNS Server

0.0.0.0

IPv6 Configuration

Automatic

Static

IPv6 Address

/

Link Local Address

FE80::20C:85FF:FE19:CEC5

Default Gateway

DNS Server

802.1X

Use 802.1X Security

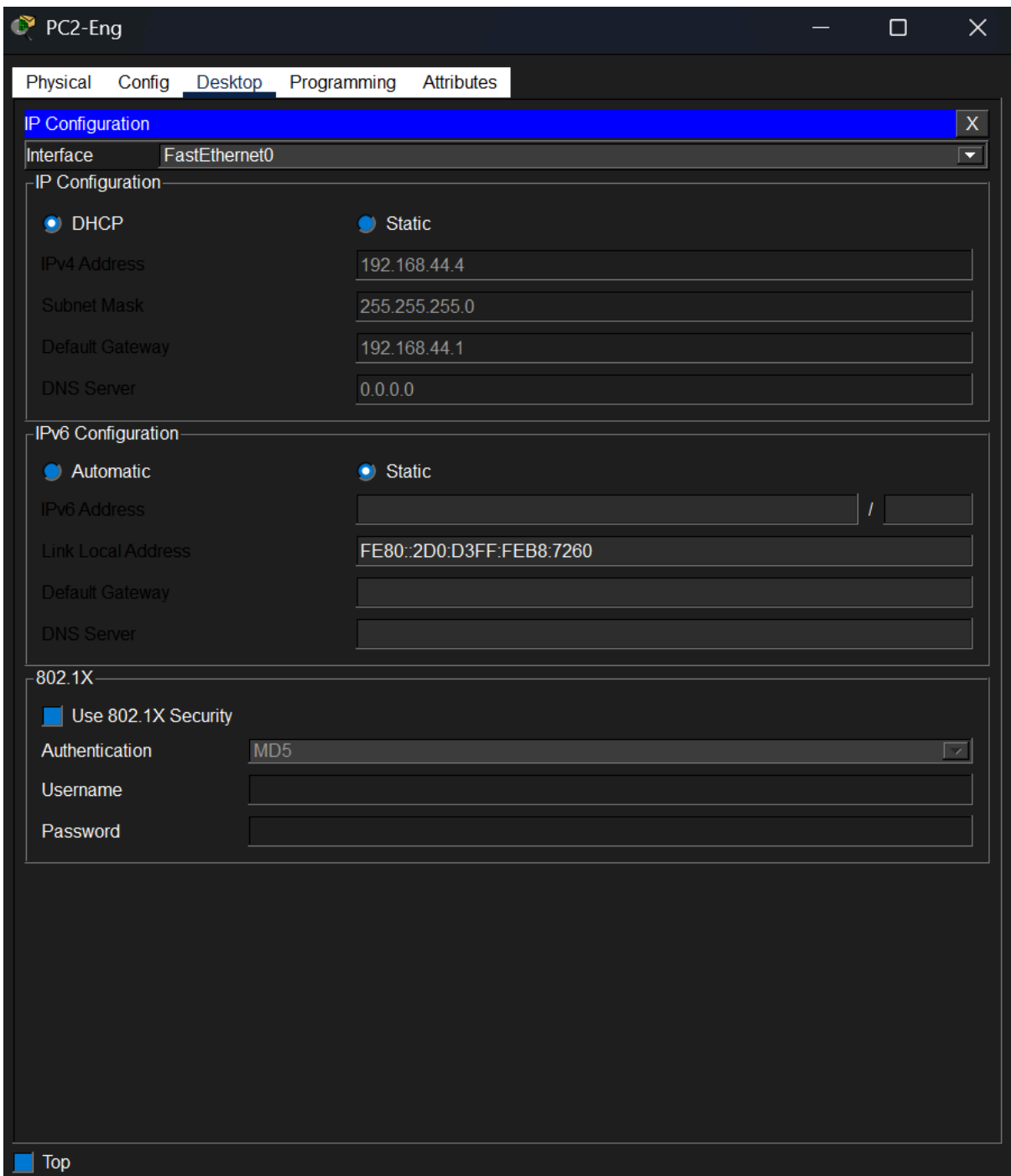
Authentication

MD5

Username

Password

Top



PC3-Eng

PhysicalConfigDesktopProgrammingAttributes

IP Configuration

X

InterfaceFastEthernet0

IP Configuration

DHCP

Static

IPv4 Address192.168.44.5

Subnet Mask255.255.255.0

Default Gateway192.168.44.1

DNS Server0.0.0.0

IPv6 Configuration

Automatic

Static

IPv6 Address/

Link Local AddressFE80::205:5EFF:FECE:1E40

Default Gateway

DNS Server

802.1X

Use 802.1X Security

AuthenticationMD5

Username

Password

Top

Server-Fin

Physical

Config

Services

Desktop

Programming

Attributes

IP Configuration

DHCP

Static

IPv4 Address

169.254.30.59

Subnet Mask

255.255.0.0

Default Gateway

0.0.0.0

DNS Server

0.0.0.0

IPv6 Configuration

Automatic

Static

IPv6 Address

/

Link Local Address

FE80::260:47FF:FE0A:1E3B

Default Gateway

DNS Server

802.1X

Use 802.1X Security

Authentication

MD5

Username

Password

Top

PC1-Fin

Physical

Config

Desktop

Programming

Attributes

IP Configuration

X

InterfaceFastEthernet0

IP Configuration

DHCP

Static

IPv4 Address169.254.76.56

Subnet Mask255.255.0.0

Default Gateway0.0.0.0

DNS Server0.0.0.0

IPv6 Configuration

Automatic

Static

IPv6 Address

Link Local AddressFE80::240:BFF:FE70:4C38

Default Gateway

DNS Server

802.1X

Use 802.1X Security

AuthenticationMD5

Username

Password

Top

PC2-Fin

Physical Config Desktop Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☒ DHCP ☒ Static

IPv4 Address 169.254.114.167

Subnet Mask 255.255.0.0

Default Gateway 0.0.0.0

DNS Server 0.0.0.0

IPv6 Configuration

☒ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::2E0:F7FF:FEC3:72A7

Default Gateway

DNS Server

802.1X

☒ Use 802.1X Security

Authentication MD5

Username

Password

Top

The end devices (PCs and Servers) across all departments were configured to obtain their network parameters. Devices within the Engineering network, such as Server-Eng and PC1-Eng, were configured with Static IP addresses (e.g., 192.168.44.2 and 192.168.44.3, respectively). Devices in

other departments were configured to request IP allocation dynamically via the DHCP protocol

PC3-Fin

PhysicalConfigDesktopProgrammingAttributes

IP Configuration

InterfaceFastEthernet0

IP Configuration

☒ DHCP

☒ Static

IPv4 Address

169.254.50.58

Subnet Mask

255.255.0.0

Default Gateway

0.0.0.0

DNS Server

0.0.0.0

IPv6 Configuration

☒ Automatic

☒ Static

IPv6 Address

/

Link Local Address

FE80::205:5EFF:FEE5:323A

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

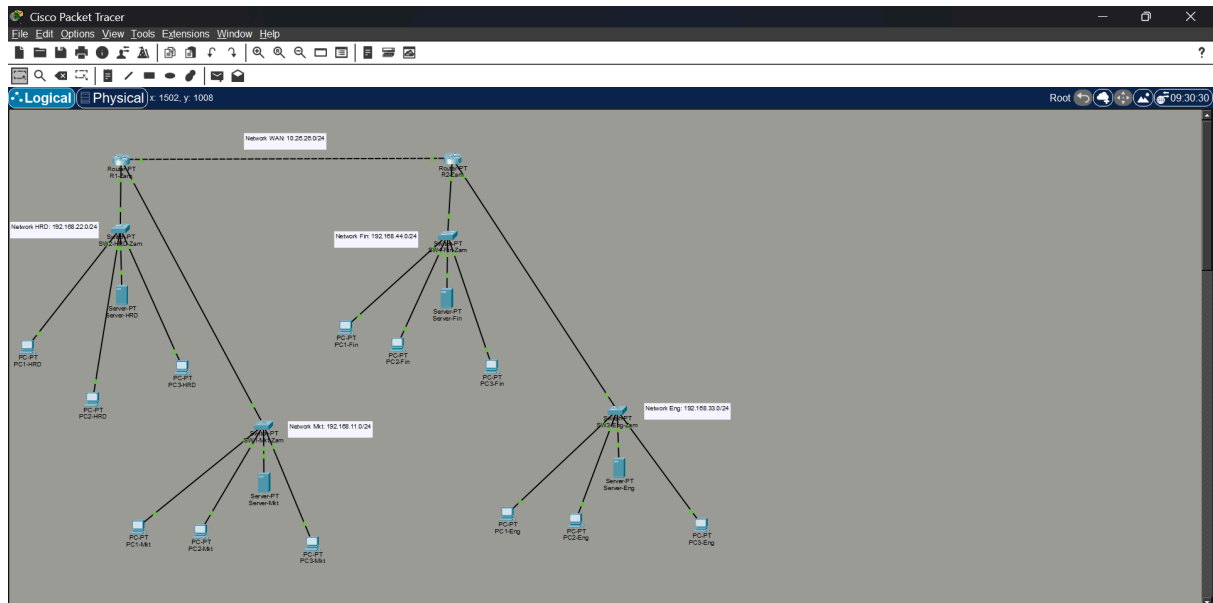
Authentication

MD5

Username

Password

Top



Conclusion

In conclusion, this Mid-Term Computer Network project successfully demonstrates the design, implementation, and configuration of a multi-departmental corporate network using Cisco Packet Tracer. By expanding the physical interfaces of the core routers with additional FastEthernet modules, the network was effectively segmented into four distinct operational areas: HRD, Marketing, Finance, and Engineering. The configuration utilized a hybrid approach to IP addressing, implementing static IPs for critical infrastructure (such as servers) and configuring DHCP pools for efficient, dynamic IP allocation across client devices. Furthermore, static routing protocols were successfully established between the core routers, enabling seamless data routing across different subnets. The final ICMP ping tests and PDU simulations confirmed 100% end-to-end connectivity, proving that all devices can communicate across the network segments efficiently and reliably, thereby fulfilling the objectives of a functional and scalable network topology.